PromoFinder: finding promo clips in microdrama screenplays

Kuku AI Generalist Assignment · Problem 1 (Find the Promos) · Om Prakash Gupta · Roll No. 230104067

1. WHY THIS PROBLEM, AND SCOPE

I picked Problem 1 because it is one editor task with a checkable output: exact timestamps plus a hook, a cliffhanger and a reason. The tool only **finds** promos for an editor to approve. It does no cutting, music or overlays, and it gives a shortlist, not a verdict.

2. ASSUMPTIONS (THE BRIEF WAS SILENT)

- Script timestamps equal the final edit. Promo length is the sum of its segments: 20–60 s, best 25–45 s.
- The 3-second hook is judged on dialogue, action and sound. Camera lines are context only.
- A promo has at most 3 segments; cross-episode clips join an episode end to the next opening.
- Spoilers are flagged, not blocked. English only. No ground-truth labels exist, so quality comes from rules and a second AI pass.

3. HOW IT WORKS: WHAT AI DOES, WHAT CODE DOES

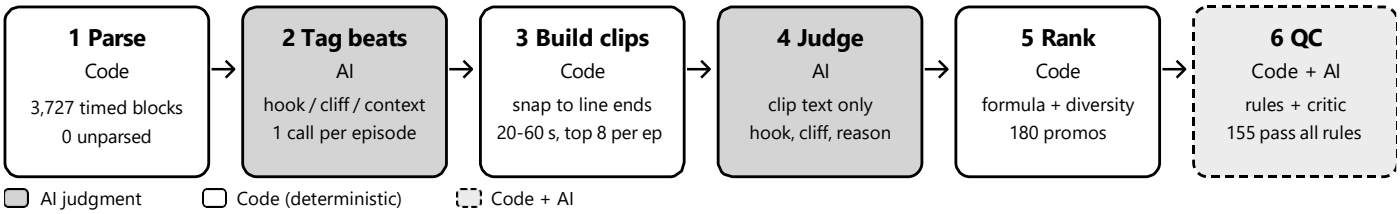


*Figure 1. Pipeline. The AI returns **beat IDs only**, never timestamps, so it cannot invent a time. Code maps IDs to times.*

Step	What happens
1 Parse	Regex reads every timed block (dialogue, action, sound, camera) and merges camera text into the line it overlaps. 0 of 3,727 blocks failed to parse across 50 files.
2 Tag	One call per episode scores each beat 0–10 for hook, cliffhanger and context needed, plus a trope tag.
3 Build	Start on a strong hook beat, end on a strong cliff beat, snap to line boundaries (never mid-line), enforce 20–60 s, drop near-duplicates, keep the top 8 per episode. Also builds 2-segment stitches and cross-episode pairs.
4 Judge	Sees **only the clip text** (like a first-time viewer) and returns the hook line, cliffhanger line, a one-line reason, a standalone score and a spoiler risk.
5 Rank	`0.30 hook3s + 0.30 cliff + 0.20 standalone + 0.15 emotion + 0.05 length-fit − spoiler penalty`, plus a diversity penalty so the top list is not one trope or one scene.
6 QC	Rule checks in code (length, hook in first 3 s, ending is not a resolution, no mid-line cut, times inside the episode) and a skeptical second-pass critic prompt.

The model is swapped with three environment variables (`LLM_PROVIDER`, `LLM_MODEL`, `LLM_API_KEY`), so a stronger model can be connected with no code change. Replies are cached, so re-runs cost nothing. Runtime model for this run: **Gemini 3.1 Flash Lite** (free tier).

4. RESULT ON THE FULL 50-EPISODE PACK

180 promos: 65 single-stretch, 103 stitched, 12 cross-episode; average length 29.5 s. The full ranked list, with hook, cliffhanger and reason for each, is in the raw outputs.

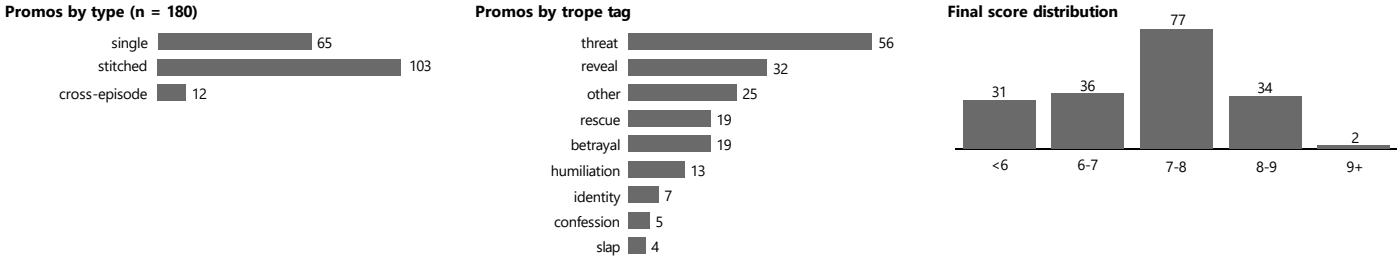


Figure 2. What was found: clip type, trope tag, and final score.

Rank	Where (episode, mm:ss)	Type · tag	Why I would post it
1	Ep 31 02:13–02:40	single · threat	Opens on a burning-book image, ends on a hot-poker close-up. Strong and self-explanatory; both ends are visual/sound text, so check on video.
2	Ep 23 01:19–01:30 + 01:58–02:11	stitched · reveal	Opens on a poisoning accusation, ends on an unresolved door-lever moment.
3	Ep 3 01:24–01:36 + 02:13–02:25	stitched · rescue	Action opener joined to an emotional, unanswered proposal.
4	Ep 36 02:16–02:28 + 03:04–03:20	stitched · threat	Opens on an execution threat with a deadline.
9	Ep 1 02:04–02:17 + Ep 2 00:00–00:12	cross-episode · betrayal	Early-funnel "previously on": the altar humiliation continued into episode 2; low spoiler risk.

I would post early-episode, low-spoiler clips first because they funnel viewers into episode 1, and hold back the 15 clips the judge marked high spoiler risk.

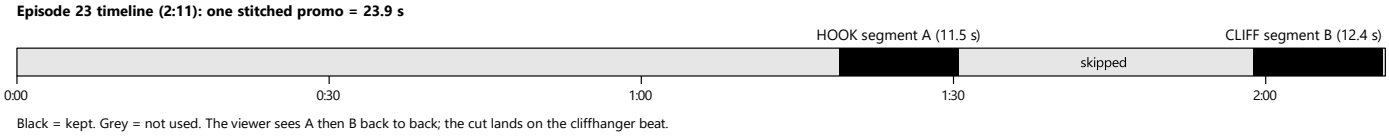


Figure 3. Illustration of a stitched promo (rank 2): two segments of one episode played back to back.

5. HOW I KNOW IT IS GOOD, AND WHERE IT FAILS

Checks run on all 180 clips

- **Rules (code):** 155 of 180 (86%) pass all five.
- **Second-pass critic:** flags 118 as confusing to a new viewer and 32 as resolving the tension, and clashes with the judge on 8. It is harsh, so it is a ranking hint, not a veto.
- **Unit tests:** 39 pass (parser, snapping, windows, ranking, QC).

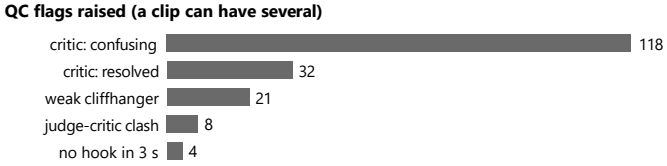


Figure 4. QC flags raised.

Where it picks badly (from the real run)

- **Sound/visual "cliffhangers":** some clips end on a sound-effect swell or camera description. The model counts these as unresolved moments; text alone cannot confirm they land on screen.
- **Missed episodes:** 5 episodes return no promo. Eps 6, 8, 17 are quiet scenes (best scores ≤4), which looks right. Ep 7 has no beat scoring 7+ as a cliffhanger. Ep 22 has hook 8 and cliff 10 yet no valid clip was built: the builder is stricter than the tagger scores suggest.
- **Stitches may feel disjoint:** 103 of 180 are stitched and none has been watched.
- **Text only:** acting, framing and audio are invisible.



Figure 5. Promos per episode. The five gaps are episodes 6, 7, 8, 17 and 22.

6. SCALE, COST AND THE HUMAN IN THE LOOP

Item	Estimate
AI calls per show	3 per episode (tag, judge, critic) + 2 per consecutive pair for cross-episode clips: about 250 for 50 episodes, well under 1M tokens.
Time	About 18 s per episode end to end in a test. My full run was slower only because the free tier rate-limited me; the client now backs off and resumes from cache.
Cost	Free tier here. I did not compute paid pricing; I would use the current price sheet.
Many shows	Per-episode work scales linearly, runs in parallel and can use batch APIs (typically about half price).
Human	An editor approves the top 10 per show; rejections become labels to tune the ranking weights and prompt examples.

7. AI TOOLS USED, AND WHERE THEY GOT IT WRONG

The runtime model is **Gemini 3.1 Flash Lite**.

Where AI got it wrong: a suggested default model, `gemini-2.5-flash`, had been retired for new users, so the first real call failed with a 404. The replacement it pointed to (`gemini-3.8-flash`) worked but took about 3 minutes per call, which was impractical, so I listed the available models and switched to the lite variant (about 18 s per episode).